Tousif Anwar

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EDUCATION

Drexel University, College of Computing & Technology — **Philadelphia, PA**
MS in Computer Science (GPA: 3.91) — *Anticipated Graduation Date: June 2026*
- Artificial Intelligence, Machine Learning & Software Architecture Specialization

Western Governors University — **Salt Lake City, UT**
BS in Computer Science (GPA: 4.00) — *Graduated: January 2024*

SKILLS

languages: Python, Java, C, C++, C#, Rust, Go, JavaScript, Typescript, SQL, LaTeX

Tools: Git, Docker, Kubernetes, REST APIs, PostgreSQL, MySQL, MongoDB, AWS, Azure, Power BI, Tableau

Frameworks: React, Node.js, Spring Framework, .NET, Vue.js

Operating Systems: Linux, Windows, macOS

RELEVANT COURSEWORK

Software Architecture	Machine Learning	Human Computer Interaction
Software Analytics	High Performance Computing	Applications of Machine Learning

EXPERIENCE

Freelance – AI Automation for Mortgage Loan Origination Firm — **Philadelphia, PA**
Software Architect — *04/2025 – Present*
- Architected an AI-powered mortgage origination platform to automate end-to-end sales pipeline workflows.
- Engineered agent runtime & cognition systems to enable intelligent task execution, contextual reasoning & decision automation across loan origination pipelines.
- Built orchestration infrastructure for multi-channel data capture and workflow execution, integrating conversational inputs and downstream processing systems.
- Leveraged LLM-based application design & distributed systems principles to deliver scalable, resilient automation.

Outlier AI — **Remote, USA**
Full Stack Software Engineer for AI Training — *01/2024 – Present*
- Designed and answered complex software engineering tasks to train and improve large language models across full-stack development, debugging, and coding best practices.
- Reviewed and benchmarked AI-generated code in JavaScript, Python, Go, Java, TypeScript, and C++ for correctness, maintainability, and production readiness.
- Analyzed architectural quality, code health, and engineering tradeoffs to ensure alignment with industry-standard software development practices.
- Provided expert guidance on Git workflows, collaborative engineering, and debugging strategies to improve model performance in real-world development scenarios.

Clear One Advantage — **Baltimore, MD**
Business Intelligence Engineer — *01/2020 - 01/2024*
- Architected and maintained scalable data models and ETL pipelines that improved data quality, consistency, and accessibility for enterprise reporting.
- Delivered high-impact dashboards and real-time business intelligence solutions that increased visibility into operational and performance metrics.
- Managed core data infrastructure for large-scale dataset storage, access, and analytics workflows.
- Collaborated with cross-functional teams to define reporting requirements, solve business problems with data, and translate technical insights into strategic recommendations.

RESEARCH EXPERIENCE

Drexel Vision Lab — **Philadelphia, PA**
Research Assistant, Computer Vision / Deep Learning — *03/2024 – Present*
- Led research on low-light pedestrian detection and image enhancement, building deep learning frameworks to improve scene visibility and detection reliability in nighttime environments.
- Developed and trained computer vision models for low-light enhancement and pedestrian detection, integrating data preprocessing, augmentation, and end-to-end evaluation pipelines in PyTorch.
- Benchmarked model performance using detection and image quality metrics, identifying key tradeoffs between perceptual enhancement quality and downstream detection effectiveness.

PROJECT

Autonomous Disaster Recovery Reporting tool – Applications of Machine Learning
- Designed an AI-powered disaster recovery platform that combined autonomous drone navigation, real-time field intelligence, and LLM-based reporting for emergency response operations.
- Built reinforcement learning–driven navigation workflows to enable drones to adapt to dynamic disaster environments and improve data collection efficiency.
- Developed a real-time reporting engine that transformed live mission and sensor data into actionable recovery updates for response teams.
- Integrated perception, reasoning, and reporting into a unified system to improve situational awareness, coordination, and disaster recovery decision-making.

CERTIFICATIONS

ITIL Service Management	AWS Solutions Architect	Artificial Intelligence
Linux Red Hat	Azure Data Engineer	GCP Machine Learning Engineer